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Name: \_\_\_\_\_

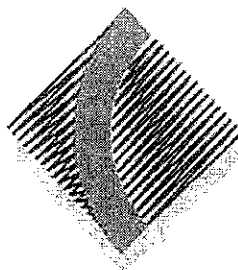
Class: 12MTZ1

Teacher: MRS GOBERT

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# CHERRYBROOK TECHNOLOGY HIGH SCHOOL



2012 AP4

**YEAR 12 TRIAL HSC EXAMINATION**

## MATHEMATICS EXTENSION 2

*Time allowed - 3 HOURS  
(Plus 5 minutes reading time)*

### Directions to candidates

- Attempt all questions
- Approved calculators may be used.
- Standard Integral Tables are provided at the back of this paper.
- Write your name and class in the space provided at the top of this question paper.

**Section I - TOTAL MARKS 10**

- To be answered on the removable answer grid at the back of the exam paper,
- Allow about 15 minutes for this section.

## Section II - TOTAL MARKS 90

- All answers to be completed on the writing paper provided. Each question is to be commenced on a new page clearly marked Question 11, Question 12, etc on the top of the page. Write your name and class at the top of each page.
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- Allow about 2 hours and 45 minutes for this section.

**YOUR ANSWERS WILL BE COLLECTED IN ONE BUNDLE. THE MULTIPLE CHOICE SECTION I ON TOP AND THEN WRITTEN ANSWERS TO SECTION II AND THEN THE QUESTION PAPER.**

## SECTION I 10 MARKS

### INSTRUCTIONS

- Attempt all questions
- Allow about 15 minutes for this section
- Section I answers are to be completed on the multiple-choice answer sheet attached to the back of this question paper.
- Select the alternative A, B, C or D that best answers the question

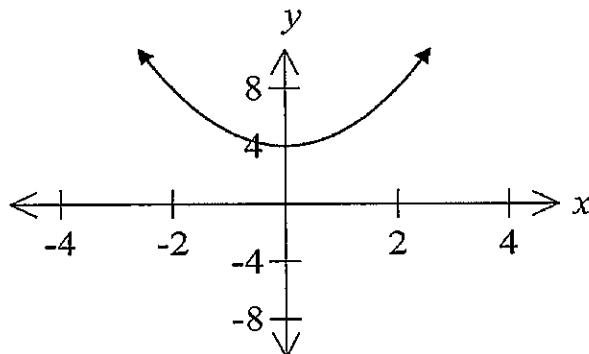
1. Find  $\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)^9$   
 (A) -1 (B) 1 (C) 0 (D) 2

2. The eccentricity of the ellipse  $\frac{x^2}{25} + \frac{y^2}{16} = 1$  is

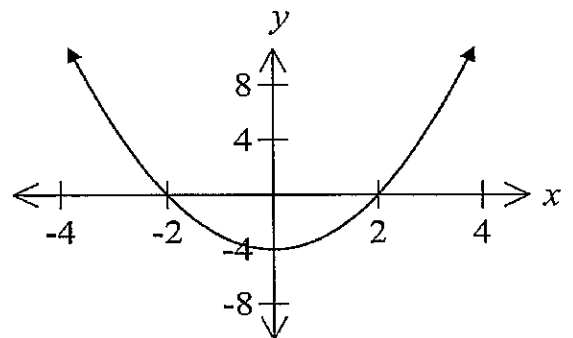
(A)  $\frac{3}{5}$  (B)  $\frac{4}{5}$  (C)  $\frac{5}{3}$  (D)  $\frac{5}{4}$

3. The graph of  $y = |f(x)|$ , given that  $f(x) = 4 - x^2$ , is

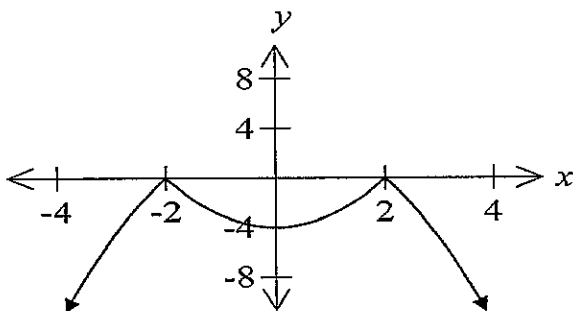
(A)



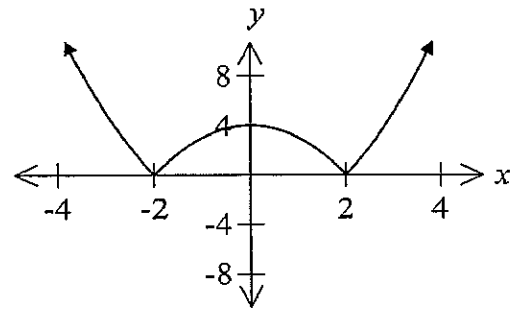
(B)



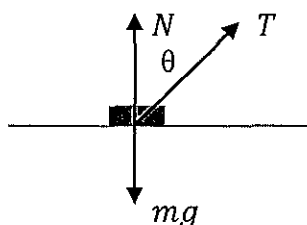
(C)



(D)



4. The parametric equations of the hyperbola  $x^2 - y^2 = 4$  are
- (A)  $x = 2 \tan \theta$  and  $y = 2 \sec \theta$     (B)  $x = 2 \sec \theta$  and  $y = 2 \tan \theta$
- (C)  $x = 4 \tan \theta$  and  $y = 4 \sec \theta$     (D)  $x = 4 \sec \theta$  and  $y = 4 \tan \theta$
5. A body of mass  $m$  kg is being pulled along a smooth horizontal table by means of a string inclined at  $\theta$  to the vertical. The diagram below indicates the forces acting on the body. Which one of the following is true?



- (A)  $N - mg = 0$
- (B)  $N + T \sin \theta - mg = 0$
- (C)  $N - T \sin \theta - mg = 0$
- (D)  $N + T \cos \theta - mg = 0$
6. The equation  $x^3 + 2x - 1 = 0$  has roots  $\alpha, \beta$  and  $\gamma$ . Find the monic equations with roots  $\alpha^2, \beta^2$  and  $\gamma^2$
- (A)  $x^3 + 2x^2 + 2x - 3 = 0$
- (B)  $x^3 + 4x^2 + 2x - 1 = 0$
- (C)  $x^3 + 4x^2 + 4x - 1 = 0$
- (D)  $4x^3 + 4x^2 + 4x - 1 = 0$
7. If  $x = \theta - \sin \theta$  and  $y = 1 - \cos \theta$ , then the expression for  $\frac{dy}{dx}$  is
- (A)  $\cot^2 \frac{\theta}{2}$     (B)  $\cot \frac{\theta}{2}$     (C)  $\tan \frac{\theta}{2}$     (D)  $\tan^2 \frac{\theta}{2}$
8. Find  $\int \frac{1}{x^2 + 2x + 2} dx$
- (A)  $\tan^{-1}(x+2) + c$
- (B)  $\tan^{-1}(x+1) + c$
- (C)  $\sin^{-1}(x+1) + c$
- (D)  $\cos^{-1}(x+1) + c$

9. Which expression is equal to  $\int 3\sqrt{x} \ln x \, dx$

(A)  $2x\sqrt{x}\left(\ln x - \frac{2}{3}\right) + c$

(B)  $2x\sqrt{x}\left(\ln x + \frac{2}{3}\right) + c$

(C)  $\frac{1}{\sqrt{x}}\left(\frac{3}{2}\ln x - 1\right) + c$

(D)  $\frac{1}{\sqrt{x}}\left(\frac{3}{2}\ln x + 1\right) + c$

10. What is the solution to the inequation  $\frac{x(5-x)}{x-4} \geq -3$ ?

(A)  $2 \leq x < 4$  or  $x \geq 6$

(B)  $1 \leq x < 4$  or  $x \geq 5$

(C)  $4 < x \leq 6$  or  $x \leq 2$

(D)  $4 < x \leq 5$  or  $x \leq 1$

END OF SECTION I

**SECTION II 90 MARKS****INSTRUCTIONS**

- Answer all questions on the writing paper provided.
- Allow about 2 hours and 45 minutes for this section.
- Begin each question on a new page and write your name clearly.
- Show all necessary working.

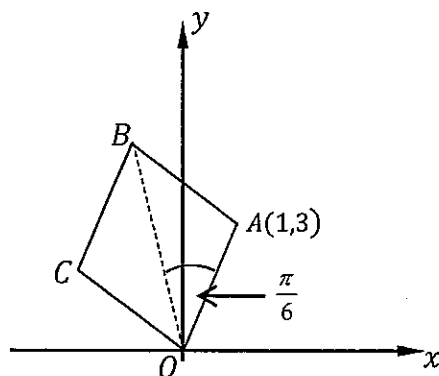
**Question 11 (15 marks) BEGIN A NEW PAGE****Marks**

- (a) Find  $\int \frac{e^{2x}}{e^x + 1} dx$ . 2
- (b) Using the substitution  $u = \pi - x$ ,
- (i) Show that  $\int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx = \int_0^\pi \frac{(\pi - x) \sin x}{1 + \cos^2 x} dx$  2
- (ii) Hence deduce that  $\int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx = \frac{\pi^2}{4}$  3
- (c) (i) Find real numbers  $a$ ,  $b$  and  $c$  such that 2
- $$\frac{8 - 2x}{(1 + x)(4 + x^2)} = \frac{a}{1 + x} + \frac{bx + c}{4 + x^2}$$
- (ii) Hence evaluate in simplest form 2
- $$\int_0^4 \frac{8 - 2x}{(1 + x)(4 + x^2)} dx$$
- (d) Let  $I_n = \int_1^e (\ln t)^n dt$  for  $n = 0, 1, 2, \dots$
- (i) Show that  $I_n = e - nI_{n-1}$  for  $n = 1, 2, 3, \dots$  2
- (ii) Hence or otherwise, find the exact value  $I_3$ . 2

(a) (i) Show that  $(1-3i)^2 = -8-6i$ . 1

(ii) Hence solve the equation  $2z^2 - 8z + (12+3i) = 0$  2

- (b) OABC is a rhombus in the Argand diagram, where O is the origin and point B is in the second quadrant. The point A is (1,3) and the angle AOB is  $\frac{\pi}{6}$ .



Copy the diagram onto your writing paper and find the complex numbers represented by the points B and C in the form  $a + ib$ . 3

(c) Given  $|z| < \frac{1}{2}$ , show that  $|(1+i)z^3 + iz| < \frac{3}{4}$ . 2

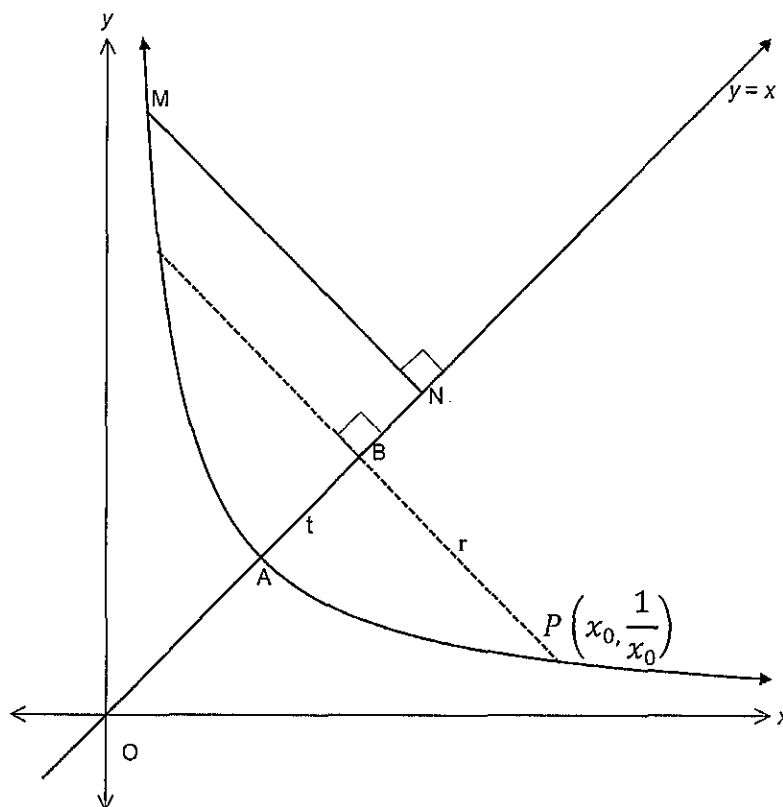
(d) Given that  $w$  is a root of  $z^3 + iz^2 + ikz + 2i = 0$ , where  $k$  is real, and  $(1-i)w$  is real. Find the possible values of  $k$ . 3

(e) Let  $z = \cos\theta + i\sin\theta$ .

(i) Show that  $z^n + \frac{1}{z^n} = 2\cos n\theta$  and find similar expression for  $z^n - \frac{1}{z^n}$ . 2

(ii) Hence prove that  $2^5 \sin^4\theta \cos^2\theta = \cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2$ . 2

(a)



The diagram shows the curve  $xy = 1$  for  $x > 0$  where the points  $A$  and  $B$  lie on the line  $y = x$ . Given  $PB$  and  $MN$  are perpendicular to  $AN$ , where  $AN = h$ , let  $AB = t$  and  $BP = r$ .

- (i) Show that  $2r^2 = \left(x_0 - \frac{1}{x_0}\right)^2$  1
- (ii) Hence show that  $OP^2 = 2(1 + r^2)$ . 1
- (iii) Show that  $r^2 = (t + \sqrt{2})^2 - 2$ . 2
- (iv) Hence, by taking slices perpendicular to  $y = x$ , of thickness  $\delta t$ , show that the volume  $V$  of the solid formed by rotating the area enclosed by the curve and the lines  $y = x$  and  $MN$ , about  $y = x$ , is given by 4

$$V = \frac{\pi h^2}{3} (h + 3\sqrt{2}).$$

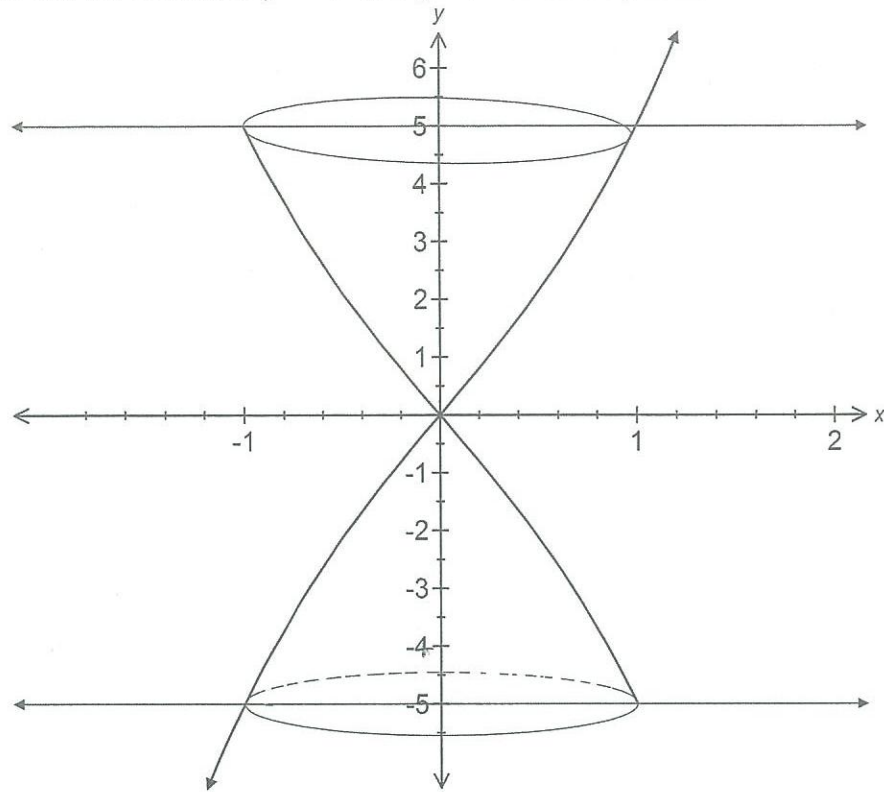
Question 13 continued next page.....

**Question 13 (Continued)**

**Marks**

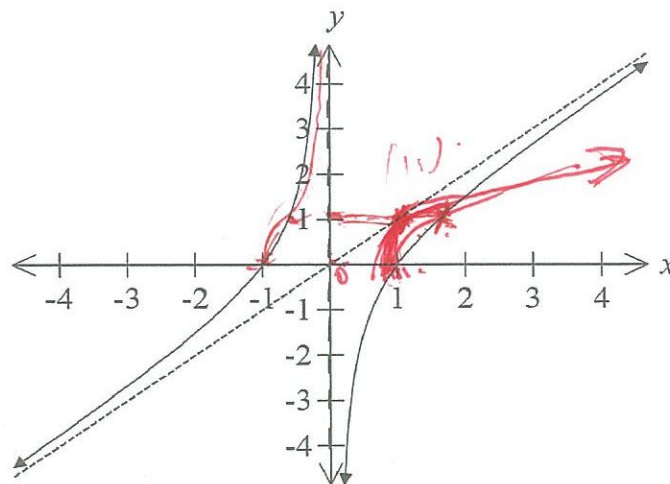
- (b) A solid is formed by rotating the region bounded by the curve  $y = x^3 + 4x$ , the  $y$ -axis and the ordinates  $y = -5$  and  $y = 5$  about the  $y$ -axis.

**5**



By using the method of cylindrical shells, find the volume of this solid.

- (c)



Copy the graph of  $y = f(x)$  shown above.

On the same number plane draw the graph of  $y^2 = f(x)$ .

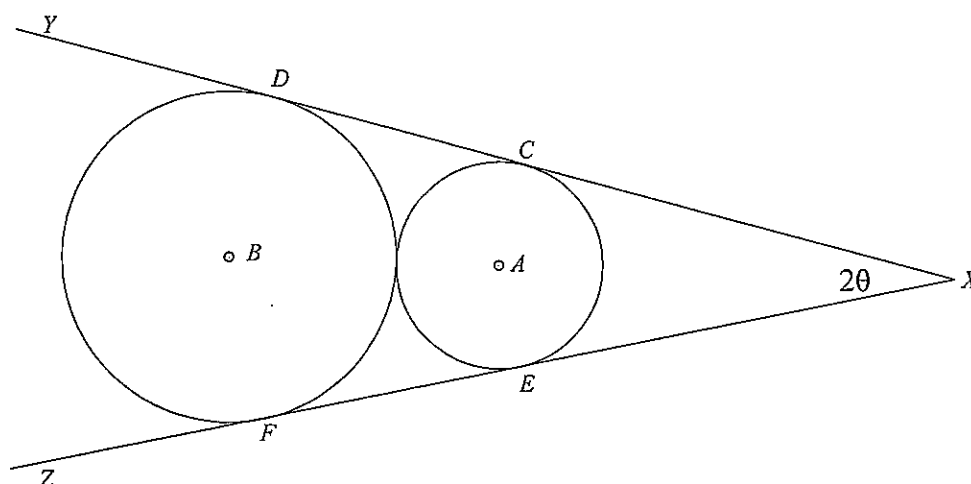
**2**

**Question 14** (15 marks) BEGIN A NEW PAGE

**Marks**

- (a) (i) Given  $x$  is real and  $x > 0$  show that  $x + \frac{1}{x} \geq 2$ . 1
- (ii) Hence show by Mathematical Induction that 3
- $$x^n + \frac{1}{x^n} \geq x^{n-1} + \frac{1}{x^{n-1}} \quad \text{for all positive integers } n \geq 1.$$

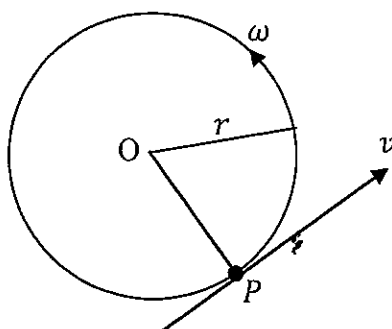
- (b) Two circles with centres  $A$  and  $B$  are in contact with each other.  
Two straight lines  $XY$  and  $XZ$  are tangents to the circle with  $\angle YXZ = 2\theta$ .



Copy the diagram onto your writing paper and

- (i) Prove that line  $AX$  bisects  $\angle YXZ$  2
- (ii) Prove that the points  $X, A$  and  $B$  are collinear. 4
- (c) Given that
- $$\tan(\alpha + \beta + \gamma) = \frac{\tan \alpha + \tan \beta + \tan \gamma - \tan \alpha \tan \beta \tan \gamma}{1 - \tan \alpha \tan \beta - \tan \beta \tan \gamma - \tan \gamma \tan \alpha},$$
- 2
- evaluate  $\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{4} + \tan^{-1} \frac{1}{13}$ .
- (d) (i) Show that  $\sin x + \sin 3x = 2 \sin 2x \cos x$ . 1
- (ii) Hence solve  $\sin x + \sin 2x + \sin 3x = 0$  for  $0 \leq x \leq 2\pi$ . 2

- (a) A particle is projected vertically upwards under gravity in a medium where the resistance to motion is proportional to the square of its speed. The speed of projection is equal to the terminal velocity  $V$  m/s, of the particle when it falls in the same medium.
- (i) Let  $x$  be the height of the particle above its point of projection and  $v$  m/s is its velocity at time  $t$ .  
Show that  $\ddot{x} = -\frac{g}{V^2}(V^2 + v^2)$ , where  $g$  is the acceleration due to gravity. 2
- (ii) Given  $H$  is the maximum height above the point of projection, show that  $H = \frac{V^2 \ln 2}{2g}$ . 2
- (iii) Given  $T$  is the time taken to achieve maximum height, show that for  $0 \leq t \leq T$ ,  $t = \frac{V}{g} \left\{ \frac{\pi}{4} - \tan^{-1} \left( \frac{v}{V} \right) \right\}$ . 1
- (iv) Given that  $\frac{2V^2}{v^2 + V^2} = 1 + \sin \left( \frac{2g}{V} t \right)$ , show that the time for the particle to reach half its maximum height is  $\frac{V}{2g} \sin^{-1}(\sqrt{2} - 1)$  seconds. 1
- (b) The diagram below shows a particle moving in a circular path of radius  $r$  with an angular speed of  $\omega$ .



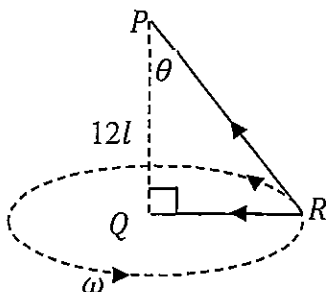
- (i) State, without proof, the tangential and normal components of the acceleration  $a$ , of the particle. 1
- (ii) What effect, if any, is there on these components if  $\omega$  is constant? 1

Question 15 continued next page

Question 15 (Continued)

Marks

- (c) The ends of a piece of string of length  $18l$  m are attached to two fixed points  $P$  and  $Q$ .  $Q$  is at a depth  $12l$  m below  $P$  and  $\angle PQR = 90^\circ$ . A small, smooth ring  $R$  of mass  $m$  kg is threaded onto the string. The ring revolves with constant speed  $\omega$  rads/sec in a horizontal circle with  $Q$  as centre and the string is taut.



- (i) Copy the diagram and indicate on your diagram the tensions in the string at  $R$ . 2  
Hence show that the tension in the rope is  $\frac{13mg}{12}$ .
- (ii) Find the period of the motion. 2
- (iii) The ring  $R$  is now **tied** to the string so that  $QR = 5l$ , and the ring 3  
revolves with constant speed in a horizontal circle with  $Q$  as centre, with  
the string remaining taut.  
Let the acceleration along  $QR$  be  $\Omega$  rads/sec.  
Copy the diagram again and indicate the active tension(s) for this situation.  
If the period of motion is  $\sqrt{2}$  times the period of the motion when the ring  
was free to move on the string, show that the ratio of the tensions along  
 $PR$  to  $QR$  in the string is 13: 4.

**Question 16 (15 marks) BEGIN A NEW PAGE****Marks**

- (a) Find the polynomial  $P(x)$ , given that it is monic, of degree 4, with -1 as a single zero and 3 as a zero of multiplicity 3. **1**
- (b) Show that when the polynomial  $f(x)$  is divided by  $(x - a)(x - b)$ , where  $a \neq b$ , the remainder is  $\frac{(x-b)f(a) - (x-a)f(b)}{a-b}$  **3**
- (c) The roots of  $z^3 - 1 = 0$  are 1,  $\Omega$  and  $\bar{\Omega}$ , where  $\Omega$  is one of the complex roots.
- (i) Explain why  $1 + \Omega + \Omega^2 = 0$  **1**
- (ii) Show that  $\bar{\Omega} = \Omega^2$  **2**
- (d) The tangent to the rectangular hyperbola  $xy = 4$  at the point  $P\left(2t, \frac{2}{t}\right)$  has equation  $x + t^2y - 4t = 0$ .
- (i) The tangent at  $P$  intersects the x-axis at  $Q(4t, 0)$  **1**  
Show that the equation of the line through  $Q$ , perpendicular to the tangent at  $P$  is  $t^2x - y - 4t^3 = 0$
- (ii) The line of equation in part (i) cuts the rectangular hyperbola at the points  $R$  and  $S$ . **2**  
Show that the midpoint  $M$  of  $RS$  is  $(2t, -2t^3)$ .
- (iii) Hence find the equation of the locus of  $M$  as  $P$  moves on the rectangular hyperbola. State any restrictions. **2**
- (e) (i) Show that  $\frac{x}{e} > \log_e x$  for  $x > e$ . **2**
- (ii) Hence explain why  $e^\pi > \pi^e$ . **1**

END OF PAPER

NAME: \_\_\_\_\_

**CTHS AP4 EXTENSION 2 MATHEMATICS 2012**

**SECTION 1 10 marks**

**Attempt Questions 1–10**

**Allow about 15 minutes for this section**

Select the correct response, A, B, C, or D . Fill in the response circle completely.

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- |    |                         |                         |                         |                         |
|----|-------------------------|-------------------------|-------------------------|-------------------------|
| 1  | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |
| 2  | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |
| 3  | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |
| 4  | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |
| 5  | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |
| 6  | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |
| 7  | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |
| 8  | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |
| 9  | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |
| 10 | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |

Markers use only

|        |      |
|--------|------|
| Q1-Q10 | /10  |
| Q11    |      |
| Q12    |      |
| Q13    |      |
| Q14    |      |
| Q15    |      |
| Q16    |      |
| TOTAL  | /100 |

# Solutions AP4 2012

## Section 1

$$\begin{aligned} \text{Q1 A } & \left( \text{cis } \frac{\pi}{3} \right)^9 \\ &= \text{cis } \left( \frac{\pi}{3} \times 9 \right) \\ &= \text{cis } \frac{\pi}{3} \\ &= \cos \pi + i \sin \pi \\ &= -1. \end{aligned}$$

$$\text{Q2. A } a=5, b=4.$$

$$\begin{aligned} b^2 &= a^2(1-e^2) \\ 16 &= 25(1-e^2) \\ e^2 &= 1 - \frac{16}{25} \\ &= \frac{9}{25} \\ e &= \frac{3}{5} \end{aligned}$$

$$\text{Q3 D}$$

$$\begin{aligned} \text{Q4 B. } x^2 - y^2 &= 4 \\ \frac{x^2}{4} - \frac{y^2}{4} &= 1 \\ a=2, b=2 \end{aligned}$$

$$\begin{aligned} x &= 2 \sec \theta, y = \tan \theta \\ -\pi < \theta < \pi, \theta &\neq \pm \frac{\pi}{2} \end{aligned}$$

$$\text{Q5 D}$$

$$\begin{aligned} \text{Q6. C } (\sqrt{x})^3 + 2(\sqrt{x}) - 1 &= 0 \\ x\sqrt{x} + 2\sqrt{x} &= 1 \\ [\sqrt{x}(x+2)]^2 &= 1 \\ x(x^2+4x+4) &= 1 \\ x^3 + 4x^2 + 4x - 1 &= 0 \end{aligned}$$

$$\begin{aligned} \text{Q7 B } x &= \theta - \sin \theta \\ \frac{dx}{d\theta} &= 1 - \cos \theta \end{aligned}$$

$$\begin{aligned} y &= 1 - \cos \theta \\ \frac{dy}{d\theta} &= \sin \theta \end{aligned}$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{\sin \theta}{1 - \cos \theta} \\ &= \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \sin^2 \frac{\theta}{2}} \\ &= \cot \frac{\theta}{2} \end{aligned}$$

$$\text{Q8 B } x^2 + 2x + 2 = (x+1)^2 + 1$$

$$\begin{aligned} \text{let } x+1 &= u \\ dx &= du \end{aligned}$$

$$\begin{aligned} \int \frac{1}{x^2+2x+2} dx &= \int \frac{1}{(x+1)^2+1} dx \\ &= \int \frac{du}{u^2+1} \end{aligned}$$

$$= \tan^{-1} u + C$$

$$= \tan^{-1}(x+1) + C$$

$$\begin{aligned} \text{Q9. A } & \int 3\sqrt{x} \ln x dx \\ &= 3 \int \ln x \frac{d}{dx} \left( \frac{2x}{3} \right)^{\frac{1}{2}} dx \\ &= 3 \left[ \frac{2x^{\frac{3}{2}}}{3} \ln - \frac{2}{3} \int x^{\frac{3}{2}} \frac{1}{x} dx \right] \\ &= 2x^{\frac{3}{2}} \ln x - 2 \times \frac{2}{3} x^{\frac{3}{2}} + C \\ &= 2x\sqrt{x} \left( \ln x - \frac{2}{3} \right) + C \end{aligned}$$

Q10. C

$$\frac{(x-4)^2 x(5-x)}{x-4} \geq -3(x-4)^2$$

$$x(x-4)(5-x) \geq -3(x-4)^2$$

$$x(x-4)(5-x) + 3(x-4)^2 \geq 0$$

$$(x-4)[x(5-x) + 3(x-4)] \geq 0$$

$$(x-4)(5x - x^2 + 3x - 12) \geq 0$$

$$(x-4)(-x^2 + 8x - 12) \geq 0$$

$$(x-4)(x^2 - 8x + 12) \geq 0$$

$$(4-x)(x-6)(x-2) \geq 0$$

~~$4 < x \leq 6$  or  $x \leq 2$~~

~~$4 < x \leq 6$  or  $x \leq 2$~~

Q11

$$(a) \int \frac{e^x}{e^x + 1} dx$$

$$= \int \frac{e^x(e^x + 1) - e^x}{e^{2x} + 1} dx$$

$$= \int \left( e^x - \frac{e^x}{e^x + 1} \right) dx$$

$$= e^x - \ln(e^x + 1) + C$$

(b) When  $x = \pi - u$

$$(i) dx = -du$$

$$= \int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx$$

$$= \int_0^\pi \frac{(\pi - u) \sin(\pi - u)}{1 + \cos^2(\pi - u)} - du$$

$$= \int_0^\pi \frac{(\pi - u) \sin u}{1 + (-\cos u)^2} du$$

$$= \int_0^\pi \frac{(\pi - u) \sin u}{1 + \cos^2 u} du$$

$$= \int_0^\pi \frac{(\pi - x) \sin x}{1 + \cos^2 x} dx$$

$$(ii) \int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx$$

$$= \frac{1}{2} \left[ \int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx + \int_0^\pi \frac{(\pi - x) \sin x}{1 + \cos^2 x} dx \right]$$

$$= \frac{1}{2} \left[ \int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx \right]$$

$$= \frac{1}{2} \left[ -\pi \tan^{-1}(\cos x) \right]_0^\pi$$

$$= \frac{1}{2} \left[ -\frac{\pi}{4} - \frac{\pi}{4} \right] = \frac{1}{2} \times \frac{\pi^2}{2}$$

$$= \frac{\pi^2}{4}$$

$$(c) \frac{8-2x}{(1+x)(4+x^2)} = \frac{a}{1+x} + \frac{bx+c}{4+x^2}$$

$$(i) 8-2x = a(4+x^2) + (bx+c)(1+x)$$

$$\text{Let } x=1, 10=5a$$

$$\therefore a=2$$

$$\text{Let } x=0, 8=2(4+0) + C(1+0)$$

$$\therefore C=0$$

Equating coefficients of  $x^2$

$$0 = 2 + b$$

$$\therefore b = -2$$

$$\therefore a=2, b=-2, c=0$$

$$(ii) \int_0^4 \left[ \frac{2}{1+x} + \frac{-2x}{4+x^2} \right] dx$$

$$= \left[ 2 \ln(1+x) - \ln(4+x^2) \right]_0^4$$

$$= 2 \ln 5 - \ln 20 - (2 \ln 1 - \ln 4)$$

$$= 2(\ln 5 - \ln 1) - (\ln 20 - \ln 4)$$

$$= \ln 5$$

$$\begin{aligned}
 \text{d) (i) } I_n &= \int_1^e (\ln t)^n dt \\
 &= \left[ t (\ln t)^n \right]_1^e - \int_1^e t n (\ln t)^{n-1} \frac{1}{t} dt \\
 &= e (\ln e)^n - 1 (\ln 1)^n - n \int_1^e (\ln t)^{n-1} dt \\
 &= e - 0 - n I_{n-1} \\
 &= e - n I_{n-1} \\
 n &= 1, 2, 3, \dots
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) } I_0 &= \int_0^e (\ln t)^0 dt \\
 &= [t]_0^e = e - 1
 \end{aligned}$$

$$\begin{aligned}
 I_1 &= e - I_0 \\
 &= e - (e - 1) \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 I_2 &= e - 2I_1 \\
 &= e - 2
 \end{aligned}$$

$$\begin{aligned}
 I_3 &= e - 3I_2 \\
 &= e - 3(e - 2) \\
 &= -2e + 6
 \end{aligned}$$

Q12.

$$\begin{aligned}
 \text{(a) (i) } (1-3i)^2 &= 1 - 6i + 9i^2 \\
 &= 1 - 6i - 9 \\
 &= -8 - 6i
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) } z &= \frac{8 \pm \sqrt{64 - 4 \times 2(12 + 3i)}}{4} \\
 &= \frac{8 \pm \sqrt{64 - 96 - 24i}}{4} \\
 &= \frac{8 \pm \sqrt{-32 - 24i}}{4} \\
 &= \frac{8 \pm 2\sqrt{-8 - 6i}}{4} \\
 &= \frac{4 \pm (1-3i)}{2}
 \end{aligned}$$

$$\text{b) } C = \left( \cos \frac{\pi}{3} \right) (1+3i)$$

$$\begin{aligned}
 &= \left( \frac{1}{2} + \frac{\sqrt{3}}{2}i \right) (1+3i) \\
 &= \frac{1}{2} + \frac{3}{2}i + \frac{\sqrt{3}}{2}i + \frac{3\sqrt{3}}{2}i^2 \\
 &= \frac{1-3\sqrt{3}}{2} + i \left( \frac{3+\sqrt{3}}{2} \right)
 \end{aligned}$$

$$\begin{aligned}
 \vec{OB} &= \vec{OA} + \vec{OB} \\
 &= (1+3i) + \left( \frac{1-3\sqrt{3}}{2} \right) + i \left( \frac{3+\sqrt{3}}{2} \right) \\
 &= \frac{3-3\sqrt{3}}{2} + \left( \frac{3+3+\sqrt{3}}{2} \right) i \\
 &= \frac{3-3\sqrt{3}}{2} + \frac{9+\sqrt{3}}{2} i
 \end{aligned}$$

(C) By  $\Delta$  inequality property

$$| (1+i)z^3 + iz | \leq | (1+i)z^3 | + | iz |$$

$$\text{LHS} = |1+i||z|^3 + |i||z|$$

$$< \sqrt{2} \left(\frac{1}{2}\right)^3 + \frac{1}{2} \quad \text{since } |z| < \frac{1}{2}$$

$$< \frac{\sqrt{2}}{8} + \frac{1}{2}$$

$$< \frac{2}{8} + \frac{4}{8}$$

$$< \frac{3}{4}$$

$$\therefore | (1+i)z^3 + iz | < \frac{3}{4}$$

d) Let  $w = x + iy$

$$(1-i)w = (1-i)(x+iy)$$

$$= x+y + i(y-x)$$

is real when  $y = x$ .

$$\therefore w = (1+i)x$$

$$w^2 = 2ix^2$$

$$w^3 = (2i-2)x^3$$

$$(2i-2)x^3 - 2x^2 + ik(1+i)x + 2i = 0$$

Since  $k$  and  $x$  are real

Equating real & imaginary parts

$$2x^3 + 2x^2 + k = 0$$

$$2x^3 + kx + 2 = 0$$

$$\therefore 2x^2 = 2$$

$$x = \pm 1$$

When  $x = 1$ ,  $k = -4$

$x = -1$ ,  $k = 0$

$$(1-i)w = 2x$$

$$w = \frac{2x}{1-i} \times \frac{1+i}{1+i}$$

$$= (1-i)x$$

(e)

(i)  $z = \cos \theta + i \sin \theta$

$$\frac{1}{z} = \cos \theta - i \sin \theta$$

$$z^n + \frac{1}{z^n} = \cos n\theta + i \sin n\theta$$

$$+ \cos n\theta - i \sin n\theta$$

$$= 2 \cos n\theta$$

$$\text{ii) } \sin^4 \theta \cos^2 \theta = \sin^4 \theta (1 - \sin^2 \theta)$$

$$= \sin^4 \theta - \sin^6 \theta$$

=

$$z^n + \frac{1}{z^n} = 2i \sin n\theta$$

ii)  $\sin^4 \theta \cos^2 \theta$

$$= \sin^4 \theta (1 - \sin^2 \theta)$$

$$= \sin^4 \theta - \sin^6 \theta$$

$$= \frac{1}{2^4} \left( z - \frac{1}{z} \right)^4 - \frac{1}{2^6} \left( z - \frac{1}{z} \right)^6$$

$$= \frac{1}{2^4} \left[ z^4 - 4z^2 + 6 - \frac{4}{z^2} + \frac{1}{z^4} \right]$$

$$+ \frac{1}{2^6} \left[ z^6 - 6z^4 + 15z^2 - 20 + \frac{15}{z^2} - 6\frac{1}{z^4} + \frac{1}{z^6} \right]$$

$$= \frac{1}{2^4} \left[ \left( z^4 + \frac{1}{z^4} \right) - 4 \left( z^2 + \frac{1}{z^2} \right) + 6 \right]$$

$$+ \frac{1}{2^6} \left[ \left( z^6 + \frac{1}{z^6} \right) - 6 \left( z^4 + \frac{1}{z^4} \right) + 15 \left( z^2 + \frac{1}{z^2} \right) - 20 \right]$$

$$= \frac{1}{2^4} (\cos 4\theta - 4 \cos 2\theta + 3)$$

$$+ \frac{1}{2^6} (\cos 6\theta - 6 \cos 4\theta + 15 \cos 2\theta - 10)$$

$$= \frac{1}{2^5} (\cos 6\theta - 2 \cos 4\theta - \cos 2\theta - 2)$$

$$2^5 \sin^4 \theta \cos^2 \theta$$

$$= \cos 6\theta - 2 \cos 4\theta - \cos 2\theta - 2$$

(5)

Q13 (a)

(i) Eqn of AN is  $x - y = 0$

$$PB = \frac{|x_0 - \frac{1}{x_0}|}{\sqrt{1+1}}$$

$$r^2 = \frac{(x_0 - \frac{1}{x_0})^2}{2}$$

$$2r^2 = (x_0 - \frac{1}{x_0})^2$$

$$\begin{aligned} \text{(ii)} \quad OP^2 &= x_0^2 + \frac{1}{x_0^2} \\ &= (x_0 - \frac{1}{x_0})^2 + 2 \\ &= 2r^2 + 2 \\ &= 2(r^2 + 1) \end{aligned}$$

iii)  $\triangle OPB$  is right angled at B

$$OP^2 = r^2 + (OA + AB)^2$$

$$OA = \sqrt{2}, \text{ since } A(1,1)$$

$$AB = t$$

$$\therefore OP^2 = r^2 + (t + \sqrt{2})^2$$

$$2(r^2 + 1) = r^2 + (t + \sqrt{2})^2$$

$$r^2 = (t + \sqrt{2})^2 - 2$$

iv) radius =  $r$ , distance  $t$  from A

thickness of disc =  $\delta t$

$$\delta V = \pi r^2 \delta t$$

$$= \pi [(t + \sqrt{2})^2 - 2] \delta t$$

$$V = \lim_{\delta t \rightarrow 0} \sum_0^h \pi [(t + \sqrt{2})^2 - 2] \delta t$$

$$= \pi \int_0^h [(t + \sqrt{2})^2 - 2] dt$$

$$= \pi \left[ \frac{1}{3} (t + \sqrt{2})^3 - 2t \right]_0^h$$

$$= \pi \left[ \frac{1}{3} (h + \sqrt{2})^3 - 2h - \frac{1}{3} (\sqrt{2})^3 \right]$$

$$= \frac{\pi}{3} [h^3 + 3h^2\sqrt{2} + 3h(\sqrt{2})^2 + (\sqrt{2})^3 - 6h - 2\sqrt{2}]$$

$$= \frac{\pi}{3} [h^3 + 3\sqrt{2}h^2]$$

$$= \frac{\pi}{3} h^2 (h + 3\sqrt{2})$$

b) Inner radius  $x$ , outer radius  $x + \delta x$

height  $5 - y$

Volume of shell

$$\delta V = \pi [(x + \delta x)^2 - x^2] (5 - y)$$

$$= \pi [2x \delta x + \delta x^2] (5 - y)$$

$$= \pi (2x + \delta x) (5 - x^3 - 4x) \delta x$$

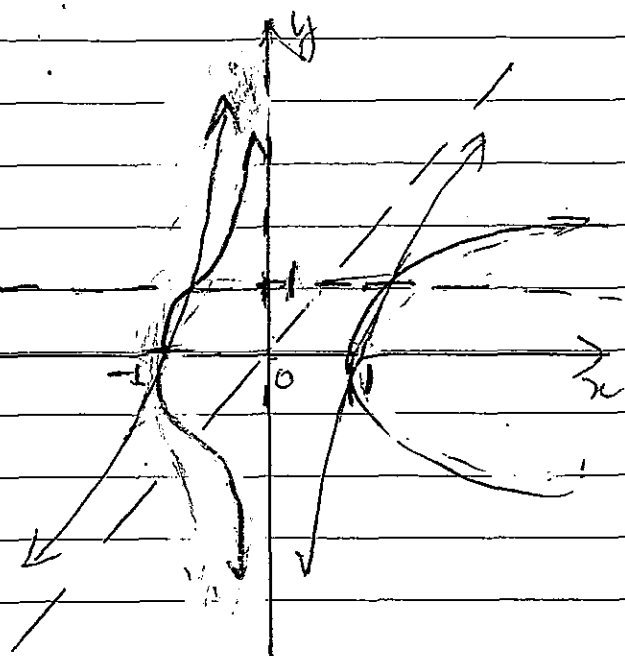
$$= 2\pi \lim_{\delta x \rightarrow 0} \sum_0^1 \pi (2x + \delta x) (5 - x^3 - 4x) \delta x$$

$$= 2\pi \int_0^1 2x (5 - x^3 - 4x) dx$$

$$= 4\pi \int_0^1 (5x - x^4 - 4x^2) dx$$

$$= 4\pi \left[ \frac{5x^2}{2} - \frac{x^5}{5} - \frac{4x^3}{3} \right]_0^1$$

$$= \frac{58\pi}{15}$$



(14)

$$ax \left( \sqrt{x} - \frac{1}{\sqrt{x}} \right)^2 \geq 0$$

$$\text{But } x + \frac{1}{x} = \left( \sqrt{x} - \frac{1}{\sqrt{x}} \right)^2 + 2$$

$$\text{for } x > 0, \quad x + \frac{1}{x} \geq 2 \quad (1).$$

(ii) When  $n=1$ ,

$$x^1 + \frac{1}{x^1} \geq x^0 + \frac{1}{x^0} \text{ by (i).}$$

$$\geq 2$$

$\therefore$  Statement true for  $n=1$

Assume statement is true for  $n=k$

$$x^k + \frac{1}{x^k} \geq x^{k-1} + \frac{1}{x^{k-1}}$$

$$\left( x^k + \frac{1}{x^k} \right) - \left( x^{k-1} + \frac{1}{x^{k-1}} \right) \geq 0$$

RTP statement is true when  
 $n=k+1$ .

Prove that

$$x^{k+1} + \frac{1}{x^{k+1}} \geq x^k + \frac{1}{x^k}$$

$$\begin{aligned} \text{LHS} &= \left( x + \frac{1}{x} \right) \left( x^k + \frac{1}{x^k} \right) - \left( x^{k-1} + \frac{1}{x^{k-1}} \right) \\ &\geq 2 \left( x^k + \frac{1}{x^k} \right) - \left( x^{k-1} + \frac{1}{x^{k-1}} \right) \\ &\quad \text{by (i)} \end{aligned}$$

$$\begin{aligned} &= \left( x^k + \frac{1}{x^k} \right) + \left[ \left( x^k + \frac{1}{x^k} \right) - \left( x^{k-1} + \frac{1}{x^{k-1}} \right) \right] \\ &\geq x^k + \frac{1}{x^k} \end{aligned}$$

$$\text{Since } \left[ \left( x^k + \frac{1}{x^k} \right) - \left( x^{k-1} + \frac{1}{x^{k-1}} \right) \right] \geq 0 \text{ by assumption.}$$

$\therefore$  Statement is true by MI

14(b)

(i) In  $\triangle AXC$  and  $\triangle AXE$

$AC = CE$  (Radii of circle)

$XC = XE$  (Tangents to circle from external point are equal)

$XA$  is common

$\therefore \triangle AXC \equiv \triangle AXE$  (SSS)

$\angle CXA = \angle AXE$  (Corresponding  $\angle$ 's in congruent  $\triangle$ 's)

$AX$  bisects  $\angle CXE$

$\therefore AX$  bisects  $\angle YXZ$  (same angle as  $\angle CXE$ )

(ii)  $\angle XCA = \angle ACD = \angle BDC = 90^\circ$   
( $\angle$  between tangent and radius at point of contact =  $90^\circ$ )

$\angle CXA = \frac{1}{2} \times \angle CXE$  (by (i)).  
Let  $\angle CXA = \theta$

$\angle CXA + \angle XCA + \angle XAC = 180^\circ$   
( $\angle$  sum of  $\triangle$ )

$\theta + 90 + \angle XAC = 180^\circ$

$\angle XAC = 90^\circ - \theta$

In  $\triangle BXD$  and  $\triangle BXF$

$DB = BF$  (radii of circle)

$BX = BX$  (Tangents from external point are equal)

$BX$  is common

$\therefore \triangle BXD \equiv \triangle BXF$  (SSS)

$\angle DXB = \frac{1}{2} \angle DXF$  (by (i))

$\angle DXB + \angle XDB + \angle XBD = 180^\circ$   
( $\angle$  sum of  $\triangle$ )

$\theta + 90 + \angle XBD = 90 - \theta$

In quadrilateral  $CDAB$

$\angle DCA + \angle CAB + \angle ABD + \angle CDB = 360^\circ$

( $\angle$  sum of quadrilateral)

$90 + \angle CAB + (90 - \theta) + 90 = 360$

$\angle CAB = 90 + \theta$

$\angle XAC + \angle CAB = (90 - \theta) + (90 + \theta) = 180^\circ$

$\therefore X, A, B$  are collinear

(c) Let  $\alpha = \tan^{-1} \frac{1}{2}$

$\therefore \tan \alpha = \frac{1}{2}$

$\beta = \tan^{-1} \frac{1}{4}$

$\tan \beta = \frac{1}{4}$

$\gamma = \tan^{-1} \frac{1}{13}$

$\tan \gamma = \frac{1}{13}$

$\tan(\alpha + \beta + \gamma) = 1$

$= \frac{1}{2} + \frac{1}{4} + \frac{1}{13} - \frac{1}{2} \times \frac{1}{4} \times \frac{1}{13}$

$1 - \frac{1}{2} \times \frac{1}{4} - \frac{1}{4} \times \frac{1}{13} - \frac{1}{13} \times \frac{1}{2}$

$= 1$

$\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{4} + \tan^{-1} \frac{1}{13}$

$= \tan^{-1} 1$

$= \frac{\pi}{4}$

Q1:

(i)  $\sin x + \sin 3x$

$$= \sin(2x-x) + \sin(2x+x)$$

$$= \sin 2x \cos x - \cos 2x \sin x$$

$$+ \sin 2x \cos x + \cos 2x \sin x$$

$$= 2 \sin 2x \cos x$$

(ii)  $\sin x + \sin 2x + \sin 3x = 0$

$$2 \sin 2x \cos x + \sin 2x = 0$$

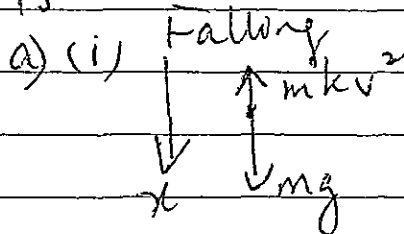
$$\sin 2x(2 \cos x + 1) = 0$$

$$\sin 2x = 0, \cos x = -\frac{1}{2}$$

$$2x = 0, \pi, 2\pi, 3\pi, 4\pi$$

$$x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi, \frac{5\pi}{2}, 3\pi, \frac{7\pi}{2}, 4\pi$$

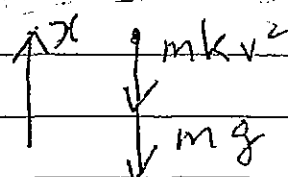
15



$$m\ddot{x} = m(g - kv^2)$$

$$\ddot{x} \rightarrow 0, v^2 = \frac{g}{k}$$

$$k = \frac{g}{V^2}$$



$$m\ddot{x} = -m(g + kv^2)$$

$$\ddot{x} = -(g + \frac{g}{V^2}v^2)$$

$$\ddot{x} = -\frac{g}{V^2}(V^2 + v^2)$$

(ii)  $\frac{1}{2} \frac{d(v^2)}{dx} = -\frac{g}{V^2}(V^2 + v^2)$

$$-\frac{2g}{V^2} \frac{dx}{d(v^2)} = \frac{1}{V^2 + v^2}$$

$$\int -\frac{2g}{V^2} dx = \int \frac{1}{V^2 + v^2} d(v^2)$$

$$-\frac{2g}{V^2} x = \ln(V^2 + v^2) + C$$

When  $t=0, x=0, v=V$

$$0 = \ln(V^2 + v^2) + C$$

$$C = -\ln 2V^2$$

$$-\frac{2g}{V^2} x = \ln(V^2 + v^2) - \ln 2V^2$$

$$-\frac{2g}{V^2} x = \ln\left(\frac{V^2 + v^2}{2V^2}\right)$$

$$x = -\frac{V^2}{2g} \ln\left(\frac{V^2 + v^2}{2V^2}\right)$$

(9)

at greatest height,  $x=H$ ,  $v=0$

$$\therefore H = \frac{-V^2}{2g} (\ln \frac{1}{2})$$
$$= \frac{V^2 \ln 2}{2g}$$

(iii)  $t \leq T$ , particle is moving upward.

$$\frac{dv}{dt} = \frac{-g}{V^2} (V^2 + v^2)$$

$$\frac{-g}{V^2} \frac{dt}{dv} = \frac{1}{V^2 + v^2}$$

$$\int \frac{-g}{V^2} dt = \int \frac{1}{V^2 + v^2} dv$$

$$\frac{-g}{V^2} t = \frac{1}{V} \tan^{-1} \left( \frac{v}{V} \right) + C$$

when  $t=0$ ,  $v=V$

$$0 = \tan^{-1} 1 + C$$

$$\therefore C = -\frac{\pi}{4}$$

$$t = \frac{V}{g} \left\{ \frac{\pi}{4} - \tan^{-1} \left( \frac{v}{V} \right) \right\}$$

for  $t \leq T$

$$(iv) \ln \left( \frac{2V^2}{V^2 + v^2} \right) = \frac{2g}{V^2} x$$

$$x = \frac{1}{2} H = \frac{1}{2} \times \frac{V^2 \ln 2}{2g}$$
$$= \frac{V^2 \ln 2}{4g}$$

$$\ln \left( \frac{2V^2}{V^2 + v^2} \right) = \frac{2g}{V^2} \times \frac{1}{2} H \ln 2$$
$$= \frac{1}{2} \ln 2$$
$$= \ln \sqrt{2}$$

$$\therefore \frac{2V^2}{V^2 + v^2} = \sqrt{2}$$

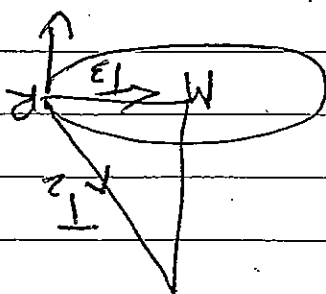
$$\therefore 1 + \sin \left( \frac{2g}{V} t \right) = \sqrt{2}$$

$$\sin \left( \frac{2g}{V} t \right) = \sqrt{2} - 1$$

$$\frac{2g}{V} t = \sin^{-1} (\sqrt{2} - 1)$$

$$t = \frac{V}{2g} \sin^{-1} (\sqrt{2} - 1)$$

iii) vertically ↓



$$T_2 = \cos \theta \, m g \quad (3)$$

$$M v_0 = m g - T_2 \cos \theta$$

$$\cos \theta = \frac{12}{13}, \sin \theta = \frac{5}{13} \quad (4)$$

$$\text{From (3)} \quad T_2 = \frac{m g}{13} = \frac{12}{13} m g$$

Period  $\sqrt{2}$  time  $T_{in (ii)}$

$$\text{Period} = \sqrt{2} \times 2\pi \frac{m}{\omega}$$

$$\therefore \frac{2\pi}{\omega} = \sqrt{2} \times 2\pi \frac{m}{\omega}$$

$$\omega = \frac{\omega}{\sqrt{2}}$$

$$= \frac{1}{\sqrt{2}} \sqrt{\frac{3g}{10L}}$$

rem (4)

$$m \times 5L \times 1.3g = T_3 + 13mg \times 5$$

$$T_3 = \frac{4}{3} m g - \frac{12}{5} m g$$

$$= \frac{m g}{3}$$

$$\therefore T_2 = T_3 = 13mg : mg$$

$$= \frac{13}{10} : \frac{1}{3}$$

$$= 13 : 4$$

111

q16.

$$(a) \, f(x) = (x+1)(x-3)^3$$

$$= (x+1)(x^3 - 9x^2 + 27x - 27)$$

$$(b) \, f(x) = T(x)(x-a)(x-b) + cx + d$$

$$f(x) = ac + d \quad (1)$$

$$f(b) = bc + d \quad (2)$$

$$(1) - (2) \quad f(a) - f(b)$$

$$= c(a-b)$$

$$\therefore c = \frac{f(a) - f(b)}{a-b}$$

Sub into (1).

$$f(a) = a \left[ \frac{f(a) - f(b)}{a-b} \right] + d$$

$$d = f(a) - a \left[ \frac{f(a) - f(b)}{a-b} \right]$$

$$= a f(a) - b f(a) + a f(b)$$

$$= a f(b) - b f(a)$$

Sub into  $cx + d$

$$\text{Remainder} = \frac{f(a) - f(b)}{a-b} x + a f(b) - b f(a)$$

$$= x f(a) - x f(b) + a f(b) - b f(a)$$

$$= (x-b) f(a) - (x-a) f(b)$$

$$a-b$$

15b)

(i) Tangential acceleration

$$\frac{dv}{dt} = \frac{d(r\omega)}{dt}$$

$$= r \frac{d\omega}{dt}$$

normal acceleration along radius

$$\text{towards } O = r\omega^2$$

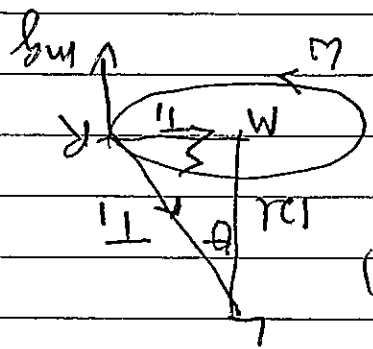
(ii)  $\omega$  is constant.

$$\frac{d\omega}{dt} = 0$$

$\therefore$  tangential acceleration = 0

normal acceleration =  $r\omega^2$

(c)(i)



Vertical forces at R downwards

$$m\ddot{x} = mg - T_1 \cos \theta$$

$$\ddot{x} = 0$$

$$\therefore T_1 \cos \theta = mg \quad (1)$$

towards M,  $MR = 5L$

$$LR = 13L$$

$$m \cdot 5L\omega^2 = T_1 + T_1 \sin \theta \quad (2)$$

$$\cos \theta = \frac{12L}{13L} = \frac{12}{13}$$

$$\sin \theta = \frac{5L}{13L} = \frac{5}{13}$$

(ii)

From (1)  $T_1 = mg \cos \theta$

$$= 13mg \times \frac{12}{13}$$

(ii) From (2)

$$m \times 5L\omega^2 = 13mg + 13mg \times \frac{12}{13}$$

$$= 18mg$$

$$\frac{12}{18mg}$$

$$\therefore \omega^2 = \frac{18mg}{12}$$

$$\frac{12m \times 5L}{18mg}$$

$$= \frac{3g}{10L}$$

$$\omega = \sqrt{\frac{3g}{10L}}$$

$$\text{Period } T = \frac{2\pi}{\omega}$$

$$= 2\pi \sqrt{\frac{10L}{3g}}$$

$$(c) \quad z^3 - 1 = 0$$

(i)  $\Omega$  is a root

$$\therefore \Omega^3 - 1 = 0$$

$$(\Omega - 1)(\Omega^2 + \Omega + 1) = 0$$

$$\therefore \Omega^2 + \Omega + 1 = 0$$

$$(ii) \quad (z - 1)(z^2 + z + 1) = 0$$

$$\text{for } z^2 + z + 1 = 0$$

$$z = \frac{-1 \pm \sqrt{1-4}}{2}$$

$$= -\frac{1}{2} \pm \frac{\sqrt{3}}{2} i$$

$$\Omega = -\frac{1}{2} + \frac{\sqrt{3}}{2} i$$

$$\bar{\Omega} = -\frac{1}{2} - \frac{\sqrt{3}}{2} i$$

$$\bar{\Omega}^2 = \left(-\frac{1}{2} - \frac{\sqrt{3}}{2} i\right)^2$$

$$= \frac{1}{4} - \frac{3}{4} - \frac{\sqrt{3}}{2} i$$

$$= -\frac{1}{2} - \frac{\sqrt{3}}{2} i$$

$$= \bar{\Omega}$$

(d) (i) grad of tangent =  $-\frac{1}{t^2}$

h grad =  $t^2$

Equation through  $Q(4t, 4t^3)$

$$y - 0 = t^2(x - 4t)$$

$$y = t^2x - 4t^3$$

$$t^2x - y - 4t^3 = 0$$

(ii)  $xy = 4$  (1)

$$t^2x - y - 4t^3 = 0$$
 (2)

$x \times (2)$

$$t^2x^2 - xy - 4xt^3 = 0$$
 (3)

Sub (1) into (3).

$$t^2x^2 - 4t^3x - 4 = 0$$

Sum of roots

$$x_1 + x_2 = -\frac{4t^3}{t^2} = -4t$$

Midpoint of RS

$$x = \frac{x_1 + x_2}{2} = \frac{-4t}{2} = -2t$$

Sub  $x = -2t$  in eqn in part (i)

$$t^2 \times -2t - y - 4t^3 = 0$$

$$y = -2t^3$$

$$M(-2t, -2t^3)$$

Alternate method solve by substitution of simultaneous equations.

16d).

(i)  $x = 2t$  (1)

$$y = -2t^2$$
 (2)

Sub  $t = \frac{x}{2}$  from (1) into

$$y = -2\left(\frac{x}{2}\right)^3$$

$$y = -\frac{x^3}{4}$$

$$4y + x^3 = 0$$

$$x^3 + 4y = 0, t \neq 0$$

16(e)

(i) Let  $f(x) = \frac{x}{e} - \log_e x$

$f'(x) = \frac{1}{e} - \frac{1}{x} > 0, x > e$

$f'(e) = 0$  and  $f(x)$  is increasing function  $x > e$

$\therefore f(x) > 0$  for  $x > e$

$\therefore \frac{x}{e} > \log_e x, x > e.$

(ii)  $\pi > e$

$\frac{\pi}{e} > \log_e \pi$  by (i)

$\pi > e \log_e \pi$   
 $= \log_e \pi^e$

$\therefore e^\pi > \pi^e.$